5.2.1 Drawing the pie layout

These are settings that need to be passed to a generator to make each of the pieces of our pie chart. This particular generator is `d3.svg.arc`, and it's instantiated like the generators we worked with in chapter 4. It has a few settings, but the only one we need for this first example is the `outerRadius()` function, which allows us to set a dynamic or fixed radius for our arcs:

Gives our arcs and resulting pie chart a radius of 100 px

```
var newArc = d3.svg.arc();
newArc.outerRadius(100);
console.log(newArc(yourPie[0]));
```

Returns the d attribute necessary to draw this arc as a <path> element: "M6.123031769111886e-15,100A100,100 0 0,1 -100,1.2246063538223773e-14L0,0Z"

Now that you know how the arc constructor works and that it works with our data, all we need to do is bind the data created by our pie layout and pass it to `<path>` elements to draw our pie chart. The pie layout is centered on the 0,0 point in the same way as a circle. If we want to draw it at the center of our canvas, we need to create a new `<g>` element to hold the `<path>` elements we'll draw and then move the `<g>` to the center of the canvas:

Appends a new <g> and moves it to the middle of the canvas so that it'll be easier to see the results

Binds the array that was created using the pie layout, not our original array or the pie layout itself

Each path drawn based on that array needs to pass through the newArc function, which sees the startAngle and endAngle attributes of the objects and produces the commensurate SVG drawing code.

```
d3.select("svg")
    .append("g")
    .attr("transform","translate(250,250)")
    .selectAll("path")
    .data(yourPie)
    .enter()
    .append("path")
    .attr("d", newArc)
    .style("fill", "blue")
    .style("opacity", .5)
    .style("stroke", "black")
    .style("stroke-width", "2px");
```

Figure 5.5 shows our pie chart. The pie chart layout, like most layouts, grows more complicated when you want to work with JSON object arrays rather than number

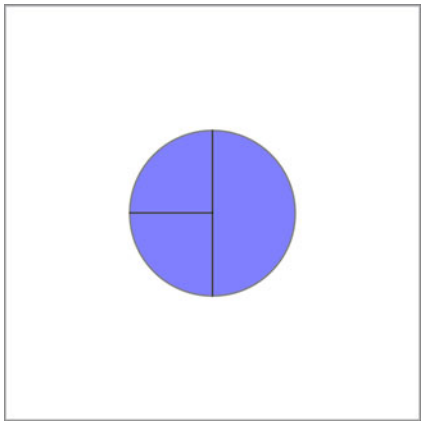


Figure 5.5 A pie chart showing three pie pieces that subdivide the circle between the values in the array [1,1,2].